

// Day 6 of 10

Day 6

10 Clean Code Rules

Developers make these every day.
Don't be one of them.

→ Swipe through all 10



Shubham Bhati

[linkedin.com/in/bhatishubham](https://www.linkedin.com/in/bhatishubham)

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Mistake #01

Meaningful Names

Code is read more than written

```
// Bad  
int d;  
String n;  
void proc(int x){ ... }
```

```
// Good  
int daysSinceCreation;  
String customerName;  
void calculateOrderTotal(int qty){ }
```



If you need a comment, rename it



Shubham Bhati

[linkedin.com/in/bhatishubham](https://www.linkedin.com/in/bhatishubham)

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Mistake #02

Functions Do ONE Thing

Single responsibility per function

```
// Bad
void processUser(User u){
    validateUser(u);
    saveToDatabase(u);
    sendWelcomeEmail(u);
    logActivity(u);
}
```

```
// Good
```



If 'and' fits in description, split it



Shubham Bhati

[linkedin.com/in/bhatishubham](https://www.linkedin.com/in/bhatishubham)

Mistake #03

Max 3 Function Params

Too many params = bad design

```
// Bad
void createOrder(String name,
    String addr, int qty, double price,
    String status, Long userId){ ... }
```

```
// Good
void createOrder(OrderRequest req){
    // All data in one clean object
}
```



Group params into a request object



Shubham Bhati

[linkedin.com/in/bhatishubham](https://www.linkedin.com/in/bhatishubham)

Mistake #04

No Magic Numbers

Numbers without context = confusing

```
// Bad
```

```
if(status == 3){ ... }  
if(age > 18){ ... }  
Thread.sleep(86400000);
```

```
// Good
```

```
if(status == ORDER_CONFIRMED){ ... }  
if(age > LEGAL_AGE){ ... }  
Thread.sleep(ONE_DAY_IN_MS);
```



Extract constants with clear names



Shubham Bhati

[linkedin.com/in/bhatishubham](https://www.linkedin.com/in/bhatishubham)

Mistake #05

DRY Principle

Duplication = multiple places to fix bugs

```
// Bad
```

```
// Same validation in 3 controllers  
if(email == null || email.isEmpty()){  
    throw new ValidationException( ... );  
}
```

```
// Good
```

```
private void validateEmail(String email){  
    if(isBlank(email)) throw new ... ;  
}
```



If you copy-paste, extract a method



Shubham Bhati

[linkedin.com/in/bhatishubham](https://www.linkedin.com/in/bhatishubham)

Mistake #06

Short Functions

If you scroll, it's too long

// Bad

```
public void doEverything(){  
    // 200 lines of code  
    // multiple responsibilities  
}
```

// Good

```
public void process(){  
    validate();  
}
```



Aim for functions under 20 lines



Shubham Bhati

[linkedin.com/in/bhatishubham](https://www.linkedin.com/in/bhatishubham)

Mistake #07

Comments Explain WHY

Code shows what, comments show why

```
// Bad
// Loop through users
for(User u : users){
    // check if active
    if(u.isActive()){ ... }
}
```

```
// Good
// Skip inactive users to avoid
```



Good code needs very few comments



Shubham Bhati

[linkedin.com/in/bhatishubham](https://www.linkedin.com/in/bhatishubham)

Guard Clauses Early

Reduce nesting with early returns

```
// Bad
void process(User u){
    if(u != null){
        if(u.isActive()){
            if(u.hasPermission()){
                // actual logic
            }
        }
    }
}
```

💡 Fail fast = flatter code



Shubham Bhati

[linkedin.com/in/bhatishubham](https://www.linkedin.com/in/bhatishubham)

Mistake #09

Consistent Formatting

Inconsistency makes code unreadable

```
// Bad
public void save(User u){if(u≠null){
repo.save(u);}
}
```

```
// Good
public void save(User u) {
    if (u = null) return;
    repo.save(u);
}
```



Use IntelliJ auto-format: Ctrl+Alt+L



Shubham Bhati

[linkedin.com/in/bhatishubham](https://www.linkedin.com/in/bhatishubham)

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Mistake #10

Tests Are Documentation

Good tests describe expected behavior

// Bad

```
@Test void test1(){ ... }  
@Test void testSomething(){ ... }
```

// Good

```
@Test  
void shouldReturn404WhenUserNotFound(){ }  
@Test  
void shouldCreateOrderForActiveUser(){ }
```



Test name = specification of behavior



Shubham Bhati

[linkedin.com/in/bhatishubham](https://www.linkedin.com/in/bhatishubham)

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done = true

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Shubham Bhati

[linkedin.com/in/bhatishubham](https://www.linkedin.com/in/bhatishubham)



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